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Paper Code : 21311

F-411

B.C.A. (Third Semester) Examination, 2024-25

(New Course)

Paper - BCA-301-N

Computer Oriented Numerical Analysis

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions. **All** questions carry equal marks. **Calculator** allowed.

1. (a) Define absolute, relative and percentage errors. An appropriate value of π is given by 3.1428571 and its true value is 3.1415926. Find the absolute and relative errors.

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(b) Solve the following:

(i) Prove that $(1+\Delta)(1-\nabla)=1$

(ii) Show that $E=(1-\nabla)^{-1}$

2. (a) Use Picard's method to obtain y for x=0.1, Given that : $\frac{dy}{dx} = x-y^2$; y=1 at x=0.

(b) Find the real root of $x^3-2x-1=0$ between 1 and 2 by bisection method. Compute five iterations.

3. (a) Find the cube root of 18 correct upto three decimal places by Regula falsi method.

(b) Evaluate $\int_3^7 x \log x dx$ taking 4 strips by simpson's ⅓ rd rule.

4. (a) Use the Runge Kutta method to approximate y for x=0.5 to x=1 using h=0.5, given that x=0 when y=1 and $\frac{dy}{dx} = \frac{1}{x+y}$

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(b) Evaluate $\int_0^6 \frac{1}{1+x^2} dx$ by simpson's 3/8th rule by dividing the interval into 6 parts.

5. (a) Find a real root of the equation $x^3-x-1=0$ using Newton Raphson method, correct to four decimal places.

(b) What are truncation errors in interpolation methods?

6. (a) Using Euler's method, find an approximate value of y corresponding to x=1, given that $\frac{dy}{dx} = x + y$ and y=1 when x=0

(b) Evaluate $I = \int_0^1 \frac{1}{1+x} dx$ correct to three decimal place using trapezoidal rule.

7. (a) Fit a straight line to the following data:

Year	1991	1992	1993	2994	1995	1996	1997	1998
Number	56	55	51	47	42	38	35	32

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(b) Find the two lines of regression for the data given below:

n=18, $\Sigma x=12$, $\Sigma y=18$, $\Sigma x^2=60$, $\Sigma y^2=96$, $\Sigma xy=48$.

8. (a) The population (in thousands) of a town in the years 1931, 1941, 1951, 1961 and 1971 are as follows:

Year	1931	1941	1951	1961	1971
Population	15	20	27	39	52

Find the population of the town in 1965 by applying Newton's backward formula.

(b) Explain Gauss Elimination methods with example.

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B.C.A. (Third Semester) Examination, 2023-24

(New Course)

Paper - BCA-301-N

COMPUTER ORIENTED NUMERICAL ANALYSIS

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions. **All** questions carry equal marks.

1. (a) Find a root of the equation $x^3-4x-9 = 0$, using bisection method correct to three decimal places.

   (b) Broadly, explain the use of Newton's divided difference interpolation formula.

2. (a) Find a real root of the equation $x^3-x-1=0$ using Newton Raphson method, correct to four decimal places.

   (b) What are truncation errors in interpolation method?

3. (a) Solve $x^3-9x+1= 0$ for the root lying between 2 and 4 by the method of false position.

   (b) Define Simpson's 1/3rd rule.

4. (a) Find solution using Newton's forward difference formula

x :	1891	1901	1911	1921	1931
f(x)	46	66	81	93	101

X= 1895

(b) Why Gauss Jordon method requires double the elementary row operations than Gauss Elimination method?

5. (a) Give comparison of Runge-Kutta and predictor corrector method.

(b) Evaluate $\sqrt{12}$ by Newton's formula.

6. (a) What is Nonlinear Regression? Comparison to Linear Regression with example.

(b) Explain Gauss elimination method with example.

7. (a) Evaluate $I = \int_0^1 \frac{1}{1+x} dx$ correct to three decimal places using trapezoidal rule.

(b) Using Euler's method, find an approximate value of y corresponding to x=1, given that $dy/dx = x + y$ and $y = 1$ when $x = 0$.

8. (a) Using Runge-Kutta method of order 4, find y for $x = 0.1$ 0.2, 0.3 given that $dy/dx = xy + y^2$, $y(0) = 1$. Find the solution at $x = 0.4$ using Milne's method.

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B. C. A. (Third Sem.) Examination, 2022-23

(New Course)

Paper No. : BCA-301-N

COMPUTER ORIENTED NUMERICAL ANALYSIS

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions. **All** questions carry equal marks.

1. (a) How many types of error? Explain any one with an example.

   (b) Describe in details numbers and their accuracy.

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2. (a) Find the absolute and relative error If $\pi=22/7$ is approximated as 3.14.

   (b) Find the real root of a equation $x^3-2x-5=0$ by the method of false position correct to three decimal places.

3. (a) Differentiate between Bisection method and Iteration method with an example.

   (b) Describe in details Newton Rapson method with an examples.

4. (a) The population of a town given below Estimate the population for the year 1925.

Year(x)	1891	1901	1911	1921	1931
Population (y)	48	68	83	95	103

   (b) Explain the rate of convergence of Iteractive methods with an example.

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5. (a) Using Newton's divided difference formula f(x) as a polynomial in x far the following data:

x	–4	–1	0	2	5
f(x)	1245	33	5	9	1335

(b) Explain the Gauss forward and back ward formula with an example.

6. (a) Use stirling's formula to find y_{35}, given $y_{20}=152$, $y_{30}=439$, $y_{40}=346$, and $y_{50}=243$.

(b) Differentiate between linear regression and line of regression.

7. (a) Use Weddle's rule to evaluate:

$$\int_0^{1/2} \frac{dx}{\sqrt{1+x^2}}$$

(b) Evaluate $\int_0^1 \frac{dx}{1+x^2}$ using:

(i) Simpson's 1/3 rule taking h=1/4

(ii) Simposon's 3/8 rule taking h=1/6

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8. (a) Differentiate between Taylor's Method and Runge-Kutta Method with an example.

(b) Explain the following statistical computation.

(i) Frequency Chart.

(ii) Data fitting with cubic splines.

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Paper Code : 21311

F-411

B.C.A. (Third Semester) Examination, 2021-22

(New Course)

Paper - BCA-301-N

Computer Oriented Numerical Analysis

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions. **All** questions carry equal marks.

1. (a) Discuss the comparison of Newton Raphson with Regula Falsi method.

   (b) Using Lagrange's interpolation formula express $(x^2+6x-1)/(x-1)(x-4)(x-6)$ as a sum of partial fractions.

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2. (a) What is Error? How to measure the accuracy of the results?

   (b) Write an algorithm to fit a regression line of Y on X by least squares.

3. (a) Given 2 1 $dy/dx=1+y^2$, with $y(0)=0$, $y(0.2)=0.20274$, $y(0.4)=0.4228$ and $y(0.6)=0.6841$. Compute $y(0.8)$ using predictor-corrector method.

   (b) Obtain a relation of the form $y=ab^x$ for the data.

X	2	3	4	5	6
Y	8.3	15.4	33.1	65.2	126.4

   By method of least squares

4. What is the difference between direct method and iterative method of find solution of non-linear equations?

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5. (a) State the following two formulae for interpolation (i) Bessel's Formula (ii) Newton's forward difference Formula.

(b) Derive formula for Newton's Forward difference interpolation.

6. (a) Obtain the smallest positive root of the equation $x^3-5x+1=0$, by using three iterations of bisection method.

(b) For solving a system of linear equations.

$a_{11}x_1+a_{12}\ x_2+a_{13}x_3=b_1$;

$a_{21}\ x_1+a_{22}\ x_2+a_{23}x_3=b_2$

and

$a_{31}\ x_1+a_{32}\ x_2+a_{33}x_3=b_3$,

by iterative Gauss–Jacobi Method, with initial approximations, $x_1 = 0 = x_2 = x_3$ give formulas for next approximations of x_1, x_2 and x_3.

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7. (a) Using Euler's method, tabulate the solution of the Initial Value Problem (IVP) $y'= -3ty^2$, $y(0) = 1$ in the interval [0, 1], using $h = 0–2$.

(b) Explain Simpson's one-third rule.

8. (a) If $f(0) = 8$, $f(1) = 68$ and $f(5) = 123$, construct a divided difference table.

(b) Difference between Lagrange Interpolation and Difference tables.

(c) give formula for trapezoidal rule.

(d) Derive dy/dx and d^2y/dx^2 from Stirling formula.

(e) Explain polynomial regression of order 2 with example?

(f) Explain the concept of pivoting.

(g) What are the various ranges of coefficient of correlation? Explain with their name.

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B.C.A. (Third Semester)

Examination, 2021

(New Course)

Paper No. BCA-301-N

COMPUTER ORIENTED NUMERICAL ANALYSIS

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions. **All** questions carry equal marks.

1. (a) Define absolute, relative and percentage errors. An appropriate value of π is given by 3.1428571 and its true value is 3.1415926. Find the absolute and relative errors.

(b) Find the real root of $x^3-2x-1=0$ between 1 and 2 by bisection method. Compute five iterations.

2. (a) Use Newton-Raphson method to find a root of the equation $x^3-3x-5=0$.

(b) Find the cube root of 18 correct upto three decimal places by Regula Falsi method.

3. (a) Evaluate $\int_0^6 \frac{1}{1+x^2}$ by Simpson's 1/3rd rule by dividing the interval into 6 parts.

(b) Evaluate $\int_3^7 x^2 \log x \, dx$ taking 4 strips by Simpson's 3/8th rule.

4. (a) Use Lagrange's formula to find f(6) from the following table

x	2	5	7	10	12
f(x)	18	180	448	1210	2028

(b) The population (in thousands) of a town in the years 1931, ..., 1971 are as follows :

Year	1931	1941	1951	1961	1971
Population	15	20	27	39	52

Find the population of the town in 1965 by applying Newton's backward formula.

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5. (a) Apply Laplace-Everett's formula to obtain the value of y_{25}, given that : $y_{20}=2854$, $y_{24}=3162$, $y_{28}=3544$, $y_{32}=3992$.

(b) Evaluate sin(0.197) using Stirling's formula from the data given below :

x	sin x
0.15	0.14944
0.17	0.16918
0.19	0.18886
0.21	0.20846
0.23	0.22798

6. (a) Explain the derivation of Lines on Regression.

(b) Solve the following :

(i) Prove that $(1+\Delta)(1-\nabla)=1$.

(ii) Prove that $E=(1-\nabla)^{-1}$.

7. (a) Use Picard's method to obtain y for x=0.1. Given that : $\frac{dy}{dx} = x - y^2$; y=1 at x=0.

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(b) Use the Runge-Kutta Method to approximate y for x=0.5 to x=1 using h=0.5 given that x=0 when y=1 and $\frac{dy}{dx} = \frac{1}{x+y}$.

8. (a) Use the Taylor's method to solve the equation $y'=x^2+y^2$ for x=0.25 given y(0)=1.

(b) Write short notes on the following :

(i) Relative Frequency and Cumulative Frequency.

(ii) A discrete variable and a continuous variable.

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F–411

B.C.A. (IIIrd Semester) Examination, 2019–20

(New Course)

COMPUTER ORIENTED NUMERICAL ANALYSIS

Paper-BCA-301-N

Time : 3 Hours] [Maximum Marks : 70

Note :- (i) Attempt any *five* questions. All questions carry equal marks.

(ii) Calculator is allowed.

1. (a) If 0.333 is the approximate value of $\frac{1}{3}$, find absolute, relative and percentage error.

   (b) Find the product of numbers 56.64 and 12.4 which are both correct to the significant digits given.

2. (a) What are various floating point representation of members ? Explain.

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   (b) Explain bisection method for root finding also. Find real root of $e^x = 3x$ by bisection method.

3. (a) Solve by iteration method :

$$\sin x = \frac{(x+1)}{(x-1)}$$

   (b) Given that :

x	1	2	3	4	5	6
$y(x)$	0	1	8	27	64	125

   Find the value of $y(2.5)$.

   (c) What do you mean by Inter Polation ? Explain.

4. (a) Write various merits and demerits of Lagrange's formula.

   (b) Evaluate :

$$\int_0^6 \frac{dx}{1+x^2} \text{ by using}$$

   (i) Simpson's one-third rule

   (ii) Simpson's three-eighth rule

   (iii) Trapezoidal's rule

   (iv) Weddle's rule

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5. (a) What do you mean by Single-step Method and Multi-step Method for numerical solution of differential equations ?

   (b) Explain Picard's method of successive approximation.

6. (a) Explain Runge-Kutta Method with a suitable example.

   (b) What is Automative Error Monitoring ?

   (c) Explain Taylor's method for I order differential equation.

7. (a) Fit a straight line to the following data regarding x as the independent variable :

x	1	2	3	4	5	6
y	1200	900	600	200	110	50

   (b) Fit an equation of the form $y = ae^{bx}$ to the following data by the method of least squares :

x	1	2	3	4
y	1.65	2.7	4.5	7.35

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8. Write any *two* short notes on the following :

   (i) Regression Analysis

   (ii) Statistical Quality Control

   (iii) Computer Arithmetic

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B.C.A. (Third Semester) Examination, 2018

(New Course)

Paper-BCA-301-N

COMPUTER ORIENTED NUMERICAL ANALYSIS

Time : 3 Hours] [Maximum Marks : 70

Note :– Attempt any *five* questions. All questions carry equal marks.

1. (a) What is Truncation Error ? Explain.

   (b) Find the absolute error in $w = xy + 2$ if $x = 2.35$, $y = 6.74$ and $z = 3.45$.

2. (a) Find a positive real root of $x \log_{10} x = 1.2$ using the bisection method.

   (b) Find a real root of the equation $x^3 - 2x - 5 = 0$ by the method of false position correct to three decimal places.

3. (a) Find a real root of $2x - \log_{10} x = 7$ correct to four decimal places using iteration method.

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(b) Find out the positive root of $x^4 - x = 10$ correct to three decimal places using Newton-Raphson method.

4. (a) From the following table, estimate the number of students who obtained marks between 40 and 45 :

Marks	No. of Students
30—40	31
40—50	42
50—60	51
60—70	35
70—80	31

(b) Find $f(22)$ from the Gauss forward formula :

x	$f(x)$
20	354
25	332
30	291
35	260
40	231
45	204

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5. (a) Given the values :

x	$f(x)$
5	150
7	392
11	1452
13	2366
17	5202

evaluate $f(9)$, using Lagrange's formula.

(b) Find the missing term in the following table using interpolation :

x	y
0	1
1	3
2	9
3	—
4	81

6. Evaluate $\int_0^6 \frac{dx}{1+x^2}$ by using :

(i) Trapezoidal rule

(ii) Simpson's 1/3 rule

(iii) Simpson's 3/8 rule

(iv) Weedle's rule and compare the results with its actual value.

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7. (a) Find the value of y for $x = 0.1$ by Picard's method, given that :

$$\frac{dy}{dx} = \frac{y-x}{y+x}, \quad y(0) = 1$$

(b) Solve $y' = x + y$, $y(0) = 1$ by Taylor's series method; Hence find the values of y at $x = 0.1$ and $x = 0.2$.

8. (a) Apply Runge-Kutta fourth order method to find an approximate value of y when $x = 0.2$ given that $dy/dx = x + y$ and $y = 1$ when $x = 0$.

(b) Explain the difference between linear and non-linear regression with proper example.

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